

Multiple Choice (40 marks)

1. Which part of the electromagnetic spectrum has the shortest wavelength?
 - (a) Gamma rays
 - (b) X-rays
 - (c) Radio waves
 - (d) Microwaves
2. According to the Solar Nebular theory what are asteroids and comets?
 - (a) They are the shattered remains of collisions between planets.
 - (b) They are chunks of rock or ice that condensed long after the planets and moons had formed.
 - (c) They are chunks of rock or ice that were expelled from planets by volcanoes.
 - (d) They are leftover planetesimals that never accreted into planets.
3. When a perfectly competitive labor market is in equilibrium,
 - (a) everyone who wants to work has the opportunity to do so
 - (b) individual firms face downward sloping labor demand curves
 - (c) unemployment can reach as high as 10-15 percent
 - (d) individual firms face upward sloping labor demand curves
4. If an scientist is stationed in the south pole during their winter, what season is it in the arctic region
 - (a) Fall
 - (b) Winter
 - (c) Summer
 - (d) Spring
5. In a lap winding dc machine number of conductors are 100 and number of parallel paths are 10. Find the average pitch
 - (a) 10
 - (b) 100
 - (c) 50
 - (d) 1
6. At 30 degress fahrenheit, water is

- (a) hot
 - (b) solid
 - (c) quenching
 - (d) gaseous
7. Which is an example of a human-made product?
- (a) cake
 - (b) tomato
 - (c) wheat
 - (d) coal
8. If a liquid disappears then what probably happened to it?
- (a) gas transformation
 - (b) solidifying
 - (c) melting
 - (d) cooling
9. A road cut shows a layer of shale on top of a layer of limestone. This indicates
- (a) the shale layer is older than the limestone layer.
 - (b) a fault occurred during the deposition of the layers.
 - (c) the environment changed between the times of deposition.
 - (d) volcanic activity occurred when these layers were deposited.
10. What characteristic is not representative of a type IIb muscle fibre?
- (a) Low oxidative capacity
 - (b) High fatigue resistance
 - (c) High motor unit strength
 - (d) Fast contractile speed
11. The human body processes and eliminates food waste using the organs of the excretory system. Which organelle performs a similar function in humans at the...
- (a) mitochondrion
 - (b) endoplasmic reticulum
 - (c) lysosome
 - (d) Golgi complex
12. When would you want a radiator the most?

- (a) winter
 - (b) spring
 - (c) fall
 - (d) summer
13. If an environment experiences a long drought
- (a) ponds may dry up and kill off the fish population
 - (b) plants will continue to flourish
 - (c) animals will experience a boom in reproduction
 - (d) tadpoles will mature faster into frogs
14. Which of the following best describes an advantage of using a mass production manufacturing system instead of a custom manufacturing system?
- (a) Customers can provide specific feedback to workers.
 - (b) Workers become skilled in all aspects of assembly.
 - (c) Goods can be easily modified for customers.
 - (d) Products can be made at a lower cost.
15. Alexander the Great was probably
- (a) bisexual
 - (b) asexual
 - (c) heterosexual
 - (d) pedosexual
16. To which of the following rights is a holder of a public corporation's cumulative preferred stock always entitled?
- (a) Conversion of the preferred stock into common stock.
 - (b) Voting rights.
 - (c) Dividend carryovers from years in which dividends were not paid.
 - (d) Guaranteed dividends.
17. the Earth rotating on its axis causes the sun to appear to move across the sky during what?
- (a) dark hours
 - (b) sunlight deficient
 - (c) night
 - (d) waking hours

18. Which changes the environment?
- (a) backyard vegetable garden
 - (b) television signals
 - (c) vacuuming carpet
 - (d) taking a nap
19. As the car approaches a hill it will become harder to
- (a) steer
 - (b) park
 - (c) push
 - (d) brake
20. According to folklore the 'jackalope' is an antlered version of what animal?
- (a) chicken
 - (b) rabbit
 - (c) moose
 - (d) snake

End of Paper